

MCU Selection Rationale

Summary

Primary recommendation: CC1354P10. Best overall balance of memory, Sub-G wireless coverage, medical IoT credibility, and a built-in sensor controller CPU that directly addresses the parallel FreeRTOS task concern.

Criteria

- Thread and/or Zigbee wireless support
- Enough memory to run a wireless stack + ~10 sensors simultaneously as parallel FreeRTOS tasks
- Low power consumption
- Compact, PCB-friendly package
- FCC approved
- Long-term vendor support

Top 3 Candidates

	CC1354P10 (Recommended)	ESP32-C6	nRF53XX
Vendor	Texas Instruments	Espressif	Nordic Semiconductor
Memory	256KB SRAM (288KB with parity disabled)	~512KB SRAM	~512KB SRAM + dedicated wireless core
Thread/Zigbee	Mature, industrial-grade stacks	Good, open-source stacks	Best-in-class stacks

Other Protocols	Sub-G, Wi-SUN, Matter, Amazon Sidewalk, mioty	WiFi6, Bluetooth	Bluetooth
Sub-G	Yes - wall/metal penetration	No	No
WiFi	No	Yes — useful for dev and OTA updates	No
Specific Optimization	Yes - sensor controller offloads sensor tasks instead	No	Yes - wireless stack runs on its own core
Power Amplifier	Integrated +20 dBm - strongest range of any candidate	Standard	Standard
Software	SimpleLink SDK + Sensor Controller Studio	Arduino, ESP-IDF, PlatformIO	nRF Connect / Zephyr RTOS
Power	Excellent	Good, but WiFi hardware adds baseline draw	Excellent
PCB	Simple layout, integrated radio - VQFN48 or VQFN64	Requires external flash chip	Integrated radio, clean layout

Medical/Industrial Track Record	Strong	Less proven in medical/industrial contexts	Widely used in medical devices
Best Case For Choosing It	Sub-G needed, or sensor controller architecture preferred	Development speed is the priority	Architectural robustness and wireless reliability are the priority

Why CC1354P10 Over CC1352XX

The CC1352XX was the original TI candidate but has been replaced by the CC1354P10, which is a direct upgrade on every relevant category.

	CC1352XX	CC1354P10
SRAM	~80KB - possibly insufficient for parallel FreeRTOS tasks	256KB (288KB without parity)
Flash	352KB	1024KB
Processor	ARM Cortex-M4	ARM Cortex-M33 - newer, more efficient
Sensor Controller	No	Yes - dedicated autonomous sensor CPU

Power Amplifier	Standard	Integrated +20 dBm high-power PA
Matter Support	No	Yes
Additional Protocols	Wi-SUN, KNX RF	Wi-SUN, Amazon Sidewalk, mioty
Sub-G	Yes	Yes
FCC Approved	Yes	Yes
Package	VQFN48	VQFN48 or VQFN64

CC1354P10 Sensor Controller

The Sensor Controller is a fully autonomous CPU built into the CC1354P10 that runs independently of the main ARM Cortex-M33. It handles sensor polling in the background and wakes the main CPU only when data is ready to transmit. This directly solves the parallel FreeRTOS task concern without any additional hardware.

Characteristic	Detail
Architecture	Fully autonomous CPU - runs independently of the main Cortex-M33
SRAM	4KB dedicated - separate from main CPU memory pool

Clock Speed	2MHz (low power) or 24MHz (active)
Power - Low Power Mode	32µA at 2MHz
Power - Active Mode	849µA at 24MHz
Power - ADC Sampling	1µA total system current at 1Hz sampling
Sensor Interfaces	Dedicated SPI-I2C digital sensor interface, 12-bit ADC, analog inputs
Programming Tool	Sensor Controller Studio - separate visual programming environment from TI
Typical Operation	Polls sensors on schedule, stores results in 4KB SRAM, interrupts main CPU when threshold crossed or buffer full
Main Benefit for MGPS	Sensor reads happen autonomously in background — main CPU sleeps most of the time, waking only to process and transmit

Sensor Compatibility

Sensor	Measurements	Interface	Compatible?	Notes
ABP2 Analog	Absolute Pressure	Analog	Likely yes	12-bit ADC and analog inputs available

ABP2 I2C	Differential Pressure	I2C	Yes	Native I2C support
BME690	Temp, Humidity, VoC	I2C or SPI	Partial	Basic temp/humidity compatible; VoC relies on Bosch BSEC algorithm - keep on main CPU
HSC	Absolute Pressure	I2C or SPI	Yes	Both interfaces supported
HTU20	Humidity, Temperature	I2C	Yes	Simple I2C read
HYT939P	Humidity, Temperature	I2C	Yes	Simple I2C read
MS580314	Absolute Pressure	SPI or I2C	Yes	Compatible; temperature compensation math may need main CPU
SDP33	Differential Pressure	I2C	Yes	Standard I2C — compatible
STS4X	Temperature	I2C	Yes	Simple I2C read
TMP117	Temperature	I2C	Yes	Simple I2C read

All sensors are compatible with the sensor controller over I2C or SPI. The BME690 is the only exception, and only partially, since basic temperature and humidity readings can still be offloaded while VoC processing stays on the main CPU.

Why Not the Others?

Eliminated — No Thread or Zigbee Support

ESP32-S3, STM32WLXX, EFR32FGXX/ZGXX, BP35C5

Eliminated — Other Reasons

MCU	Reason
CC1352XX	Direct predecessor to CC1354P10
nRF54XX	Successor to nRF53XX but too new - immature SDK, higher risk for prototype timeline
STM32WBXX	Dual-core and decent memory, but Thread/Zigbee stacks less reliable than Nordic or TI
ESP32-H2	Weaker version of ESP32-C6 - less memory, no WiFi, less mature
CC2652XX	2.4GHz-only version of CC1352XX - strictly weaker, no Sub-G
EFR32MGXX	Best-in-class Zigbee/Thread stack but no Bluetooth and narrow ecosystem

Brief Comparison: CC1354P10 vs STM32WL vs EFR32FGXX

	CC1354P10	STM32WL55	EFR32FG23
Vendor	Texas Instruments	STMicroelectronics	Silicon Labs
Processor	ARM Cortex-M33 @ 48MHz	Dual-core: Cortex-M4 @ 48MHz + Cortex-M0+ (radio)	ARM Cortex-M33 @ 78MHz
SRAM	256KB (288KB without parity)	64KB	64KB
Flash	1024KB	256KB	512KB
Sub-G	Yes - 287–1054 MHz	Yes - 150–960 MHz	Yes - Sub-GHz bands
2.4 GHz	Yes - dual band	No	No
Thread/Zigbee over Sub-G	Yes - native IEEE 802.15.4g support	No - LoRa and (G)FSK only	No - proprietary Sub-G protocols only
Thread/Zigbee at 2.4GHz	Yes	No	No
Bluetooth LE	Yes - BLE 5.3	No	Yes - BLE 5.3
Matter	Yes	No	No

Sensor Controller	Yes - dedicated autonomous CPU	No	No
Power Amplifier	Integrated +20 dBm	+22 dBm (some variants)	+20 dBm (some variants)
Sensor Interface	SPI, I2C, 12-bit ADC, analog inputs	SPI, I2C, ADC	SPI, I2C, ADC
Operating Temperature	-40°C to +105°C	-40°C to +105°C	-40°C to +85°C
FCC Approved	Yes	Yes	Yes
Package	VQFN48 or VQFN64	Various QFN	QFN32 to QFN40